Junior Makers · Class 5

# Rain-Sensing Automatic Window

**It feels the first drop and shuts itself!**

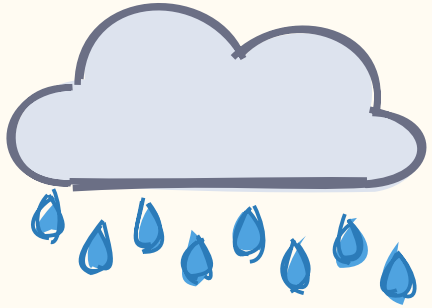
Rain sensing | Water conducts electricity | Servo hinge

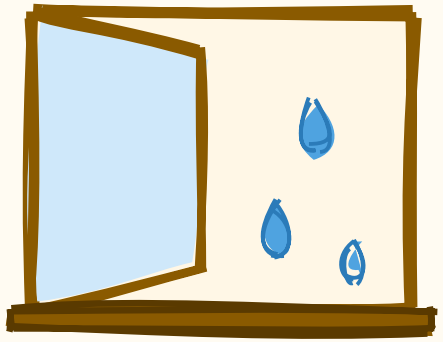


Did you know?
Many modern cars switch on their wipers by themselves when rain hits the windscreen.

Let's think...

# Monsoon rain, window open, nobody home!





**Picture this**

You are at school. The bedroom window at home is wide open. Suddenly, dark clouds and a heavy monsoon shower. Your books, bed and carpet are getting soaked. Who closes the window?

Phone a neighbour?

Run home?

The window itself!

**Today:** we build a window that feels the rain and closes itself, then warns the family.

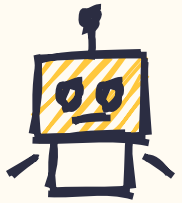
Learning aims

# Today I will learn to...

1. Explain why **water conducts electricity** and how a rain sensor uses it
2. Read a **rain sensor** and understand its numbers
3. Use a **servo** and a **hinge** to close a model window
4. Trigger an **LED** and **buzzer** alert when rain starts
5. Choose a sensible **threshold** by testing

**Keywords**

conductor · rain sensor · analog · threshold · servo · hinge · alert



Watch and think

# Video time!



**Rain-Sensing Windshield Wipers**

Toyota Canada · about 2 minutes · youtu.be/rscCDwS0OaQ

## Watch for...

1. Where is the rain **sensor** on the car?
2. What does the car do when more rain falls?
3. Why is this safer for the driver?

Scan to watch

**After the video:** Our window will close by itself when it rains. What other machine could react to rain?

Real world

# Machines that feel the rain



**Car wipers**
Switch on and speed up when rain hits the glass



**Greenhouses**
Roof vents close before plants get flooded



**Smart homes**
Skylights and windows shut and send a phone alert



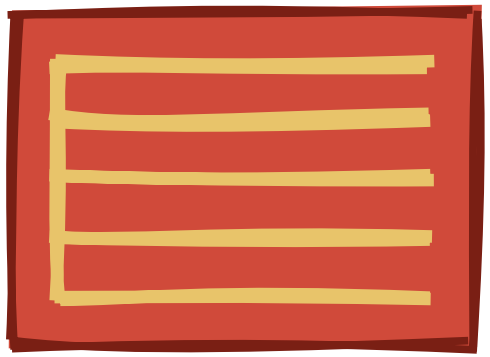
**Washing lines**
Some covers roll out over clothes when it rains

In Pakistan, the monsoon (July to September) can bring **sudden heavy rain**. A window that closes in seconds could save a room.

Spot it!
Ask at home: has rain ever come in through an open window?

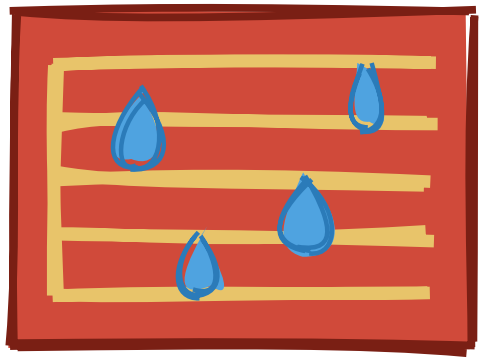
The science

# Water can carry electricity



**DRY: gap between lines**

no path, almost no current



**WET: water bridges the gap**

a path! current flows

## How the plate works

The plate has two sets of metal lines that **do not touch**.

A raindrop lands across two lines and makes a **bridge**.

More water = more bridges = more current.

### Why rain conducts

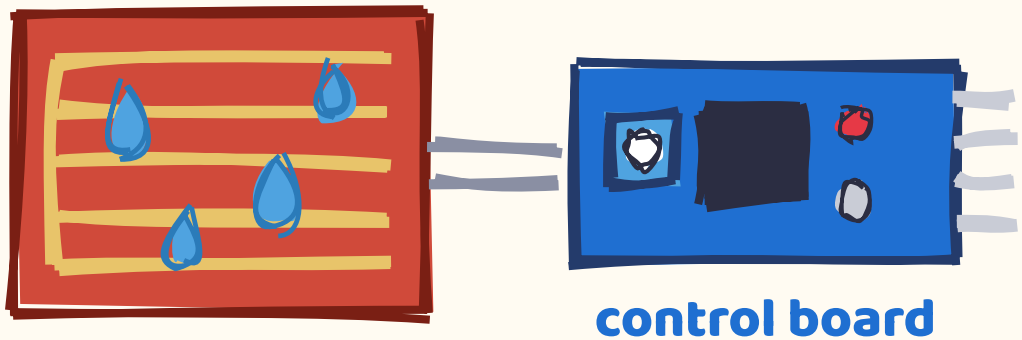
Rainwater carries tiny bits of dissolved dust and salts. Those let electricity through. Very pure water barely conducts!

**Safety:** our 5V sensor is safe. Water and **mains** electricity are deadly. Never touch switches with wet hands.

LEARN

Meet the sensor

# The rain sensor: two parts




## On the control board

**Blue dial** (potentiometer): sets when DO switches to "wet".

**Red LED**: power is on.

**Green LED**: lights up when DO says wet.

A small chip (LM393) compares the plate to the dial.

| Pin | What it does |
|---|---|
| VCC | 5V power |
| GND | Ground |
| AO | Analog out: a number 0 to 1023 (how wet) |
| DO | Digital out: LOW = wet, HIGH = dry |

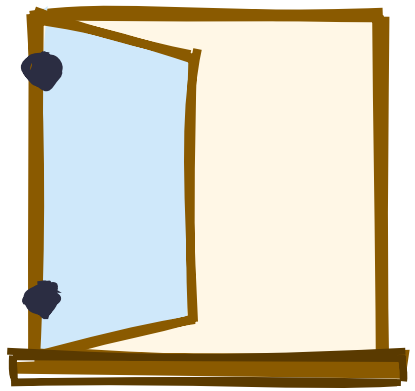
Today we use AO

A number tells us HOW wet, not just yes or no. We choose the threshold ourselves in code!

LEARN

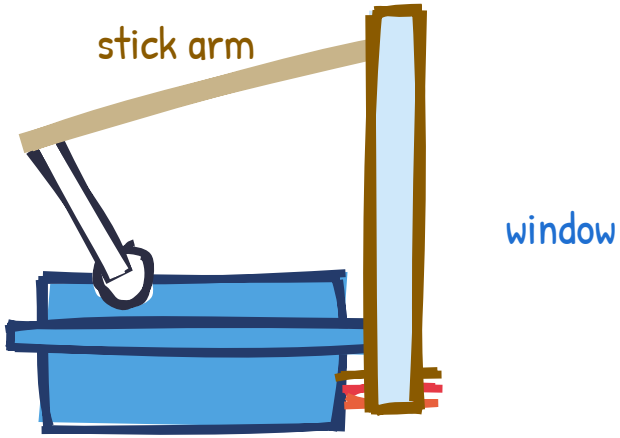
Engineering

# Hinges: how the window swings



**hinge on the side**

A **hinge** lets a door or window turn around one edge, like your elbow.




The **servo horn** pushes a **stick arm**, and the arm pushes the window shut. This is a **linkage**.

## Servo angles

**0°** = window open

**90°** = window closed

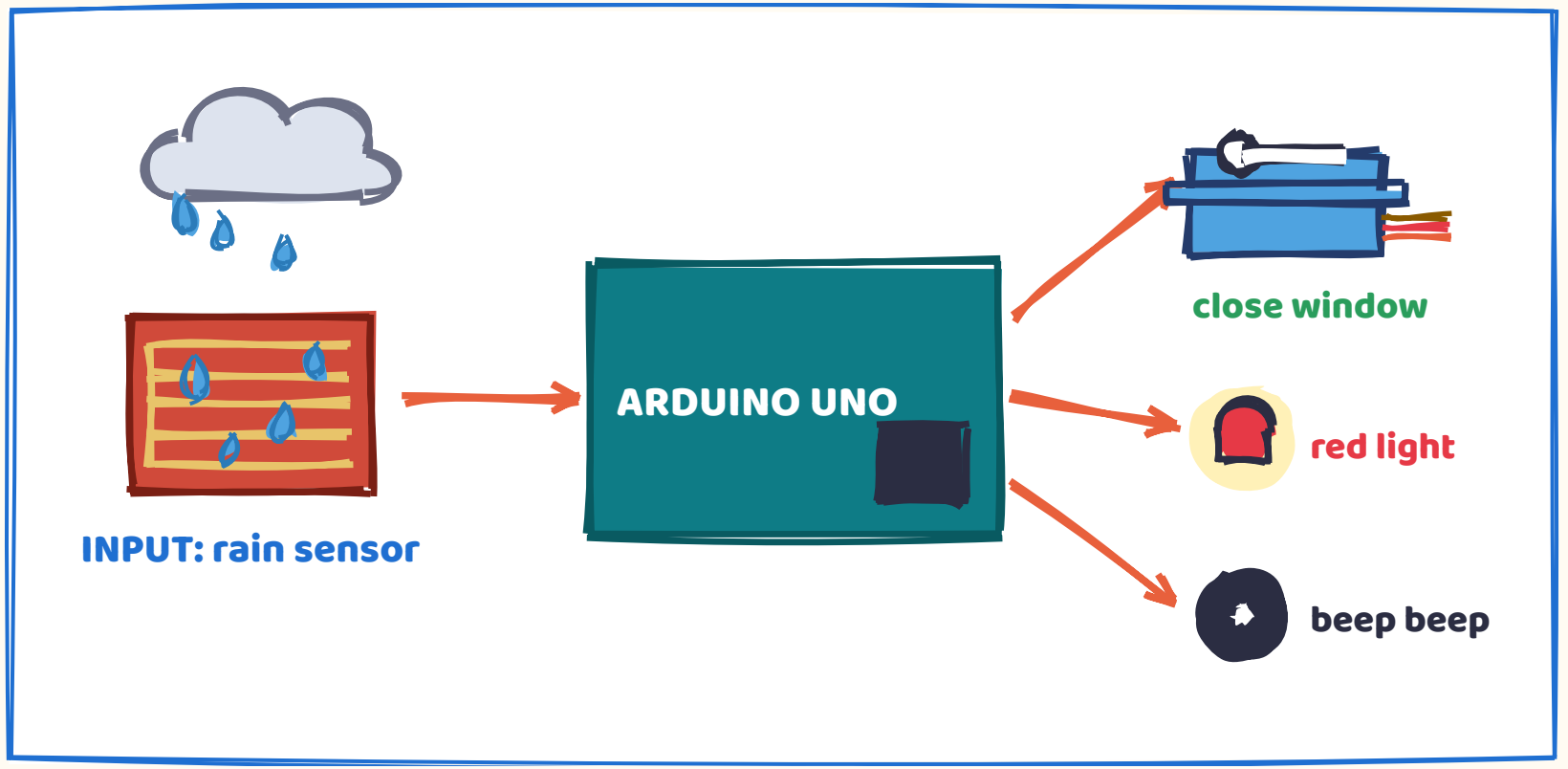
Your window may need 80° or 100°. Test it!

Hinge hack

Use a strip of strong tape along one edge. Bend it back and forth a few times to loosen it.

The plan

# One sensor, three actions



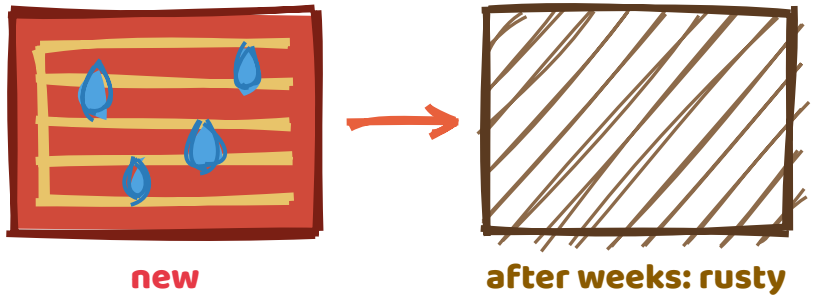

## The rule

**IF** rain number < 500
**THEN** close window, red light ON, beep.
**ELSE** keep window open, light OFF.

Remember?
Last class: two sensors, two jobs. Today: one sensor, three outputs!

Engineer's secret

# Why rain sensors wear out




When electricity flows through water on metal, the metal slowly **corrodes** (rusts away).

## How engineers fix it

Switch the sensor **ON** only when reading, then **OFF** again.

Wipe and dry the plate after class.

Real products use special coatings or light beams instead.

Teacher tip

Power the sensor VCC from a digital pin: turn it on, wait 10 ms, read, turn it off.

**End-of-class routine:** unplug USB, wipe the plate with a tissue, let it dry flat, then pack it away. Our sensors will last the whole year!

Part 2 · Make it

# Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

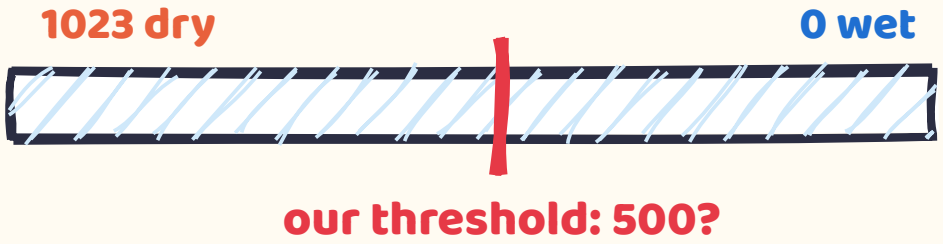
1. What number does rain make?
2. Be the rain-sensing window
3. Our window kit (per group)
4. Make the model window
5. Wiring the window
6. The rain-sensing window in ArduinoBlocks
7. Make it rain!

Be a scientist!

# What number does rain make?

## Upside-down again!

Dry plate = **big number** (near 1023).
Wet plate = **smaller number**.
Drip water on the plate with a dropper and watch the number fall.




| Plate | Typical | Ours |
|---|---|---|
| Completely dry | about 1000 | _____ |
| 2 or 3 drops | about 700 | _____ |
| Wet all over | about 400 | _____ |
| Soaked / puddle | below 300 | _____ |

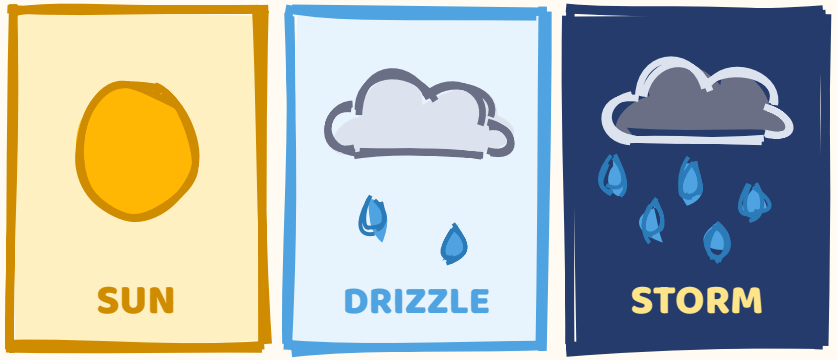
**Every sensor is different**
Typical numbers are only a guide. Test YOUR sensor and write YOUR numbers.

Stand up and play!

# Be the rain-sensing window

## How to play

1. Everyone is a window: arms **wide open**.
2. The teacher shows a card: **sun**, **drizzle** or **storm**.
3. Storm: close your arms and shout **"BEEP BEEP!"**
4. Sun: stay open. Drizzle: you decide!




? **Drizzle round:** did everyone do the same thing? That is why the machine needs **one clear number** to decide: the **threshold**.

Think deeper

Should the window close for ONE drop? What could go wrong if it did?

Build it

# Our window kit (per group)

**Arduino Uno**
The brain

**Rain sensor**
Plate + control board

**SG90 servo**
Closes the window

**Red LED + 220 Ω**
Rain alert light

**Buzzer**
Rain alert sound

**Cardboard**
Wall, frame, window

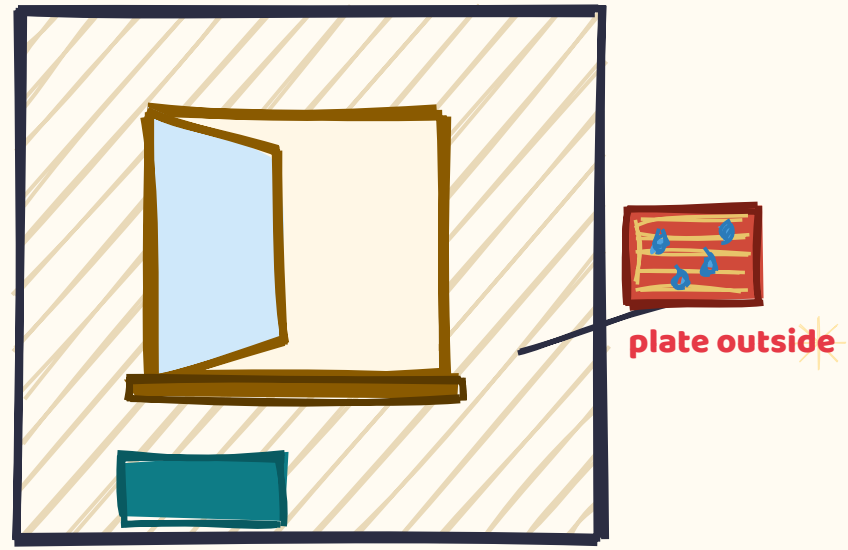
Also: jumper wires, breadboard, tape, ice-cream stick, **spray bottle or dropper**, tissues, a tray to catch drips.

Re-use
The servo and lid skills from the smart bin work here too. Engineers re-use ideas!

Build it

# Make the model window

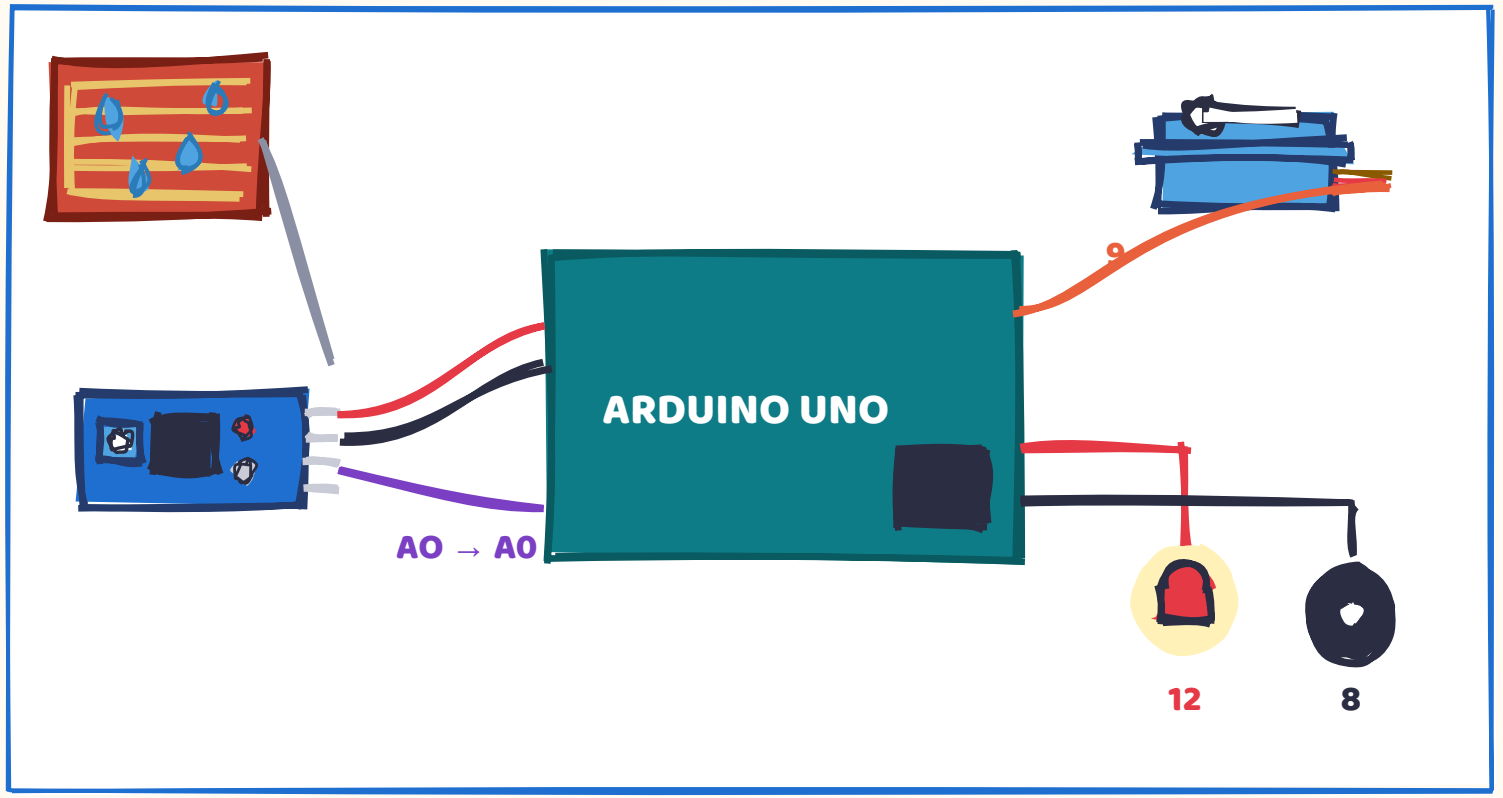
1. **Cut** a window hole in a cardboard wall. Keep the cut-out: that is your window.
2. **Hinge** the window to one side of the hole with strong tape.
3. **Fix** the servo inside the wall. Link the horn to the window with a stick.
4. **Place** the rain plate **outside**, tilted a little so water runs off.
5. **Keep** the Arduino and control board **inside**, far from the water.




**Why tilt the plate?** So the water runs off and the sensor can dry, just like a real roof.

Build it

# Wiring the window




| Part | Arduino |
|---|---|
| Sensor VCC / GND | 5V / GND |
| Sensor AO | A0 |
| Servo orange | pin 9 |
| Servo red / brown | 5V / GND |
| Red LED (+ 220 Ω) | pin 12 |
| Buzzer + | pin 8 |

**Keep it dry**
Only the rain plate gets wet. The control board, breadboard and Arduino stay inside!

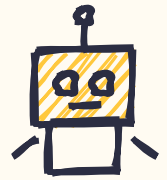
Code it

# The rain-sensing window in ArduinoBlocks

```
when Arduino starts
  servo pin 9 → angle 0 (window open)
forever
  set rain to read analog A0
  serial print rain
  if rain < 500
    servo pin 9 → angle 90 (close)
    turn ON pin 12 (red light)
    tone pin 8, 1000 Hz, 200 ms
  else
    turn OFF pin 12
  wait 500 ms
```

### Read it like a story

Start with the window **open**.
Read the rain number every half second.
**IF** it is wet: close, light up, beep.
**ELSE**: light off. The window **stays shut** until we add the challenge!



Change the 500

Use the numbers you wrote in your table. Pick a number between dry and a few drops.

Test like an engineer

# Make it rain!

| Test | Expect | Number | Result |
|---|---|---|---|
| Dry plate | Window open, quiet | _____ | pass / fail |
| 1 drop | ?? (your threshold decides) | _____ | pass / fail |
| Spray 3 times | Close, red light, beep | _____ | pass / fail |
| Wipe dry with tissue | Light off. Window? | _____ | pass / fail |

## Spray rules

Spray **only** the plate, from above, like real rain.
Never spray the Arduino, breadboard or laptop.
Keep tissues and a tray ready.

Window does not close? Open the **Serial monitor**. Is the number really going below your threshold?

Did it beep forever?
That is correct for now! The plate is still wet. Fixing it is today's challenge.

Level up!

# Window challenges

## Choose your challenge



**Sunshine again:** reopen the window when the plate is dry (number above 900).



**Beep only once:** beep 3 times when rain starts, then stay quiet.



**Rain-o-meter:** light green, yellow or red LEDs for drizzle, rain and storm.



**Gentle close:** close the window slowly, a few degrees at a time.

## Why two numbers?

Close below **500**, open above **900**. The gap stops the window flapping open and shut while the plate is half dry.

**Boss level**

Only reopen after the plate has been dry for 10 seconds. Real rain stops and starts!

Show what you know

# What did we learn?

- [x] Rainwater conducts electricity and bridges the plate's lines
- [x] Wetter plate = smaller number on A0
- [x] A threshold is the number where the machine changes its mind
- [x] A servo and a hinge make a linkage that closes the window
- [x] One sensor can trigger many outputs: motor, light and sound

**Exit ticket:** the sensor reads 320. Is the window open or closed? Why?

**At home**

Next time it rains, look for anything that reacts to the rain by itself.



**Next class:** a Fire & Gas Safety Alarm Panel!